Pavel Jelínek, Mgr.



Address: Nad Přehradou 408
109 00 Prague 15, Czech Republic
Email: pavel0jelinek@gmail.com
Phone: +420 602 443 447

English: **B2** (+ complex expression, – listening)
German: **A2** (+ conversation, – vocabulary)
Czech: **Native**

LinkedIn GitHub with a code demo

Software Developer with a strong analytical focus, mathematical background and QA mindset:

- **22 years of sw development:** Kotlin, SQL, C#, C++, ABAP, full-stack ASP.NET; Git, Claude Code.
- **SAP Business Analyst**: Data flows & integrity, migrations, impacts of changes in data or processes.
- **End-to-End Consultant**: Translating business requirements into practical solutions, covering the full lifecycle from analysis and design through implementation, testing, user training, and ongoing support.
- **QA mindset**: Attention to detail, systematic search for potential flaws, adding auto-checks everywhere.
- **Background in theoretical and applied mathematics:** exact thinking, attention to corner cases.
- **Original optimization algorithms** (routing, manufacturing) within **2%** of the best-known solutions.

Target working arrangement: 20 hours per week (4 hours per day), primarily remote. I work best in a **small team** with clearly defined, **stable assignments** and a **high degree of autonomy** allowing me to focus on **carefully thought-out solutions**.

Professional Experience

02/2020–present: Blindspot Solutions, part of Adastra Group | **Combinatorial Optimization**

- Development of the OptiSuite module for optimization using simulated annealing.
  - Improved efficiency, caching, robustness, user-friendliness, documentation, and functionality.
  - Identified numerous bugs and proactively added many missing unit tests.
- 2023–present: **Independent development of optimization algorithms.**
  - Redesigned their algorithm for the **Vehicle Routing Problem** (VRP): both the basic variant and Pickup and Delivery. Achieved results within 2% of the best-known benchmarks.
  - For manufacturing (**Job-Shop Scheduling Problem**), developed a program achieving results within 0.5% of the best-known benchmarks.
- Stack: Kotlin, IntelliJ, jUnit+Mockito, Git+BitBucket, Claude Code.

04/2019–01/2020: Cegeka | **Banking Portfolio Management App**

- Backend: SQL, C#, WebServices. GUI: WebForms, Telerik.
- Developed complex grid-layout dialog (column formatting, DB persistence).

04/2018–03/2019: Bee Interactive | **Truck Loading Management**

- Full-stack dev: SQL, C#, Razor/HTML, JavaScript, jQuery.
- ASP.NET MVC, Entity Framework, IQueryable, DatAdmin.
- Created unit tests in nUnit with Mocking libraries.

08/2005–03/2018: KCT Data | **SAP/ABAP Consultant, Analyst, Developer**

- Onboarding/mentoring of new colleagues. Internal SAP dev & consulting for HR and Accounting.
- HR/FI batch data loading; reporting across HR, SD, MM, DMS, FI, CO.
- Client projects (consulting, spec, implementation, support):
  - RolesRedesign: Streamlining user permissions.
  - Incoming Mail Book: Email/scan sorting & classification workflow.
  - Attendance Management: Clock-in/out and overtime tracking, productivity-linked bonuses.

11/2003–07/2005: Xanadu (AAC Solutions) | **C++ Programmer**

- Large-scale configuration software for PVC windows: SAP extension built on C++ and Windows MFC.
- Feature development; proactively enhanced application robustness and diagnostics.

Skills in More Detail

Analyst

- Systematic decomposition and formulation of problems, potential states, and hidden dependencies.
- Breaking complex problems into solvable parts.
- Attention to edge cases and ambiguous requirements (thanks to a mathematical background).
- Thinking through potential consequences of solutions, vulnerabilities, and the impact of future changes.

Consultant

- End-to-end responsibility from requirements gathering to implementation, testing, customer training, operation, and bug resolution.
- Experience with complex systems (SAP), data flows, and data migrations.
- Striving for comprehensive customer satisfaction. Seeing the program through the user's eyes.
- Keen awareness of unstated assumptions in requirements, potential ambiguities, and misunderstandings.

Software Developer

- Clean code, DRY, SOLID. Emphasis on documentation, clear structure, extensibility, and reliability.
- Proactive initiative to ensure overall application functionality, not just the subtask assigned to me.
    - Main languages: Kotlin, SQL, C#, C++, SAP/ABAP.
    - Limited knowledge of: ASP.NET, Python, HTML/CSS, JavaScript, assembler, PHP, Matlab.
    - xUnit, Mockito; IntelliJ, MS VS; TortoiseSVN, Git, GitHub, BitBucket; Slack, JIRA; LaTeX.
    - LLM coding assistants (Claude Code).

QA Mindset

- Active search for potential sources of bugs, unexpected behavior, or security risks.
- Data integrity and robustness: Resilience against unexpected inputs and future modifications.
- Proactive suggestions to increase robustness and self-diagnostics. Attention to detail.

Teaching Skills

- Training colleagues, clear documentation. Tutoring, including at the university level.
- Designed and led a high school programming club on computer graphics and game development.

Preferred Work Style: **Focus, autonomy and well-considered results.**

I bring the most value through independent work, clear priorities and stable assignments with sufficient autonomy to think problems through thoroughly.

I excel at solving practical technical problems and directly consulting with the end users to understand their real needs, rather than working in an environment with frequent changes in assigned tasks or excessive time spent in meetings and coordination.

Education: **Applied and theoretical mathematics, theoretical computer science.**

1999–2003 Faculty of Mathematics and Physics, Charles University in Prague

- Graduated with honors, Master's degree (Mgr.)
- Mathematical logic, abstract algebra and its applications in computer science. Computability theory.
- Master's thesis: Introduced a new type of computational automaton applicable in linguistics.

1996–1999 Faculty of Applied Sciences, University of West Bohemia in Plzeň

- Numerical methods, engineering math and mechanics, computer modelling of physical processes.
- Programming: MATLAB, Pascal, Prolog, Lisp, Ada; DOS system programming.
- "English for Engineering Professions" program.
- Individual coaching in differential equations under Ing. Brandner.

1992–1996 Mikulášské Gymnázium (Grammar School) in Plzeň

- Specialized mathematics class; achievements in programming and physics competitions.
- Senior project: Optimization using Genetic algorithms.

Personal Background

Interests: Exploring AI limits, strategic games (Chess, Go, 4X, game development), history of economics.

Hobbies: Crosswords and puzzles, fantasy, sci-fi, musical composition, improvisation, jamming.

Past Hobbies: Contributing to Wikipedia, hobby programming: games, game AI, physical simulations, computer graphics.